

Deployment Test Inventory

This document lists the automated tests and smoke checks used by `tgir_quantlib_tools` before and after deployment. It is generated from the repository workflows, runbook/deployment notes, and unittest files.

Release-Gate Tests

The DigitalOcean deployment workflow runs these unittest modules before packaging and uploading a release:

- `tests.test_cliquet`: Release-gate unittest module for cliquet pricing identities.
- `tests.test_sofr_curve`: Release-gate unittest module for SOFR curve construction, calibration, Bermudan schedules, and sensitivity behavior.
- `tests.test_smoke`: Release-gate unittest module for portfolio pricing, login/session behavior, routes, editors, APIs, and CSV download.

CI Checks

```
.venv/bin/python -m py_compile app.py wsgi.py portfolio.py tgir_quantlib_tools/*.py
```

Compiles Python sources to catch syntax/import-time parser errors.

```
.venv/bin/python -m unittest tests.test_cliquet
```

Runs cliquet identity tests.

```
.venv/bin/python -m unittest tests.test_sofr_curve
```

Runs curve, calibration, Bermudan, and sensitivity tests.

```
.venv/bin/python -m unittest tests.test_smoke
```

Runs route, session, API, editor, and CSV smoke tests.

```
make -C docs/latex
```

Builds the LaTeX documentation set used by CI.

Local Smoke Gates

```
./.venv/bin/python -m unittest discover -s tests
```

Runs all unittest modules under `tests/`.

```
./.venv/bin/python build_SOFR_curve.py
```

Builds the SOFR OIS curve, prints selected discount factors, and reports calibration-swap repricing.

```
./.venv/bin/python -c "from app import app;
    print(app.test_client().get('/').status_code)"
```

Minimal Flask route smoke check for the app entrypoint.

Post-Deploy Health Checks

```
curl -fsS http://127.0.0.1:8008/health
```

Checks the restarted WSGI service on the droplet loopback bind.

```
curl -fsS "$DO_HEALTHCHECK_URL"
```

Checks the public health endpoint through the production URL.

Individual Unittest Coverage

tests/test_cliquet.py - CliquetIdentityTests

- test_cliquet_identity_portfolio: Runs ten subcases proving QuantLib cliquet pricing reduces to simpler vanilla, forward-start strip, repeated-quarter, and zero-volatility deterministic limits.

tests/test_sofr_curve.py - SofrCurveTests

- test_curve_reprices_all_quoted_ois_swaps: Builds the SOFR curve, reprices quoted OIS calibration swaps to near-zero NPV, validates zero-rate and daily-forward rows, and checks Hull-White calibration rows.
- test_default_state_exposes_workbook_surface_and_bermudan_grid: Validates default valuation date, workbook ATM vol matrix shape, interpolation behavior, 100mm notionals, and Bermudan grid dimensions.
- test_swap_point_sensitivities_show_all_curve_nodes_and_zero_vega_cells: Checks swap point sensitivities across all SOFR nodes and confirms the swap has zero vol-matrix and model-parameter exposure.
- test_european_swaption_only_moves_selected_matrix_pillar: Confirms European swaption vega is isolated to the selected 1Y x 4Y matrix pillar and has mean-reversion sensitivity.
- test_bermudan_swaption_uses_diagonal_calibration_strip_only: Confirms Bermudan calibration uses the expected diagonal strip and reports vega only on the relevant source/interpolated pillars.
- test_workbook_bermudan_calibration_strip_matches_call_schedule: Checks the second Bermudan trade's exercise mapping, calibration labels, source market points, and probability disclosure.
- test_bermudan_schedule_keeps_fixed_final_maturity_and_shortens_underlying_tenor: Verifies a fixed final maturity schedule with shortening underlying tenors and expected exercise dates.
- test_workbook_reference_bermudan_tracks_excel_mark: Prices the workbook-reference Bermudan and checks it remains near the expected Excel mark of about 1.414mm.
- test_g2pp_bermudan_model_builds_with_two_factor_parameters: Builds the G2++ Bermudan model and verifies the expected a, sigma, b, eta, and rho parameter rows.

tests/test_smoke.py - PortfolioSmokeTests

- test_price_portfolio_returns_expected_rows: Prices the default portfolio and verifies the five expected blotter rows have NPVs.
- test_login_page_renders: Checks that the public login page renders the sign-in heading and workstation action.
- test_dashboard_requires_login: Confirms the protected dashboard redirects unauthenticated users to login.
- test_login_redirects_to_dashboard: Logs in and verifies the dashboard renders core navigation, market, curve, vol, diagnostic, and cliquet sections.
- test_market_update_round_trips: Posts curve, model, vol, and equity market inputs and verifies the updated values render back to the page.
- test_realtime_tick_perturbs_curve_and_equity_market: Calls the realtime tick API and validates updated market payloads, blotter rows, Greeks, and no NaN output.
- test_trade_editor_updates_swap_terms: Posts swap term changes and verifies the dashboard reflects the receiver direction, tenor, rate, and frequencies.

- test_default_rates_trades_start_at_100mm_notional: Confirms default rates trades start at 100mm notional and use relative price error calibration.
- test_trade_editor_updates_bermudan_frequencies_and_fixed_maturity: Updates Bermudan terms and verifies model, calibration method, schedule mapping, frequencies, and notional rendering.
- test_equity_cliquet_trade_editor_renders_analytics: Checks the equity cliquet editor renders mark, reset decomposition, scenarios, Monte Carlo profile, and SPX context.
- test_trade_editor_renders_point_sensitivities: Verifies trade editor sensitivity controls and initial zeroed analytics state.
- test_trade_risk_api_returns_sensitivity_payload: Calls the trade risk API and validates base NPV, curve rows, and full vega matrix dimensions.
- test_trade_template_falls_back_when_trade_risk_missing: Renders the trade template without risk context and verifies graceful fallback output.
- test_health_is_public: Checks /health is public and returns the expected JSON payload.
- test_quantlib_data_model_page_renders: Verifies the QuantLib data-model page renders curve, instrument, model, engine, cliquet, and research content.
- test_ibkr_compare_dashboard_renders: Checks the IBKR compare dashboard renders blotter, Greeks, totals, market panels, grid, cliquet assumptions, and diagnostics.
- test_curve_debug_csv_downloads: Downloads curve-debug CSV and validates headers, date coverage, populated SOFR/zero rows, final forward blank, and file persistence.

Coverage Summary

Together these checks cover QuantLib SOFR curve calibration, Bermudan model calibration and schedules, cliquet pricing identities, portfolio pricing, Flask login/session flows, protected routes, trade editors, realtime and risk APIs, debug CSV export, Python compilation, documentation build, and production health endpoints.